

Modern Algebra I, Math 6310  
University of Utah, Fall 2026  
Gordan Savin

Class MWF 12:55 – 1:45 pm, LCB 323

Office hours LCB 205, MWF 9:40 – 10:30 am, and by appointment.

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**Goals** This is the first course in the graduate algebra sequence; the main topics are Rings, Modules, and Commutative Rings. These roughly correspond to Parts II, III and V, respectively, in *Abstract Algebra* by Dummit and Foote.

**Texts** *Abstract Algebra* by Dummit and Foote.

**Prerequisites** C or better in Math 5310

**Homework** Problem sets will be posted and updated every other week on Canvas. The homework will not be collected or graded but will be discussed in class.

**Exams** We will have three in class exams based on the problem sets. Each exam will follow a chapter from the book and will deal with the topics from that chapter. Approximate grading scale: C for 50% or better, B for 65% or better, A for 80% or better.

**Final Exam** Monday, December 14, 1:00-3:00 in the classroom. This is a comprehensive exam with 5 problems. The exam is mandatory for graduate students, and optional for undergraduate students. Two correct solutions count as a *pass*, and three correct as a *high pass*. A high pass will also be given to students scoring A during the term and with two correct solutions on the final exam.

**Other dates** Last day to add, drop, audit, or elect CR/NC: Friday, September 04. Last day to withdraw from classes: Friday, November 13.

**Outcomes/objectives** Students should master the following mathematical notions and theorems, to be used in the subsequent research: Ideals, quotient rings, group rings, matrix rings, division rings, commutative rings, prime and maximal ideals, group of units, principal ideal domains, unique factorization domains, structure theorem for modules over a principal ideal domain and its applications to abelian groups and to linear algebra, rational and Jordan forms, eigenvectors, eigenvalues, minimal and characteristic polynomials, Cayley-Hamilton Theorem. Free modules, submodules, quotient modules, tensor, symmetric and exterior products of modules, projective and injective modules, exact sequences, complexes, homology, connecting homomorphisms, Tor and Ext functors. Polynomial rings, Gauss' Lemma, Eisenstein's criterion, localization and field of fractions,

noetherian rings, Hilbert's Basis Theorem, integral extensions, Noether's Normalization Lemma, Hilbert's Nullstellensatz.

### Information, policies, and resources

**The Americans with Disabilities Act:** The University of Utah seeks to provide equal access to its programs, services, and activities, for people with disabilities. If you will need accommodations in the class, reasonable prior notice needs to be given to the Center for Disability & Access, 162 Olpin Union Building, (801) 581-5020, <https://disability.utah.edu/>. CDA will work with you and the instructor to plan for accommodations. All written information in this course can be made available in alternative format with prior notification to the Center for Disability & Access. In compliance with ADA requirements, some students may need to record course content. Any recordings of course content are for personal use only, should not be shared, and should never be made publicly available. In addition, recordings must be destroyed at the conclusion of the course.

**Safety at the U:** The University of Utah values the safety of all campus community members. You will receive important emergency alerts and safety messages regarding campus safety via text message. For more information regarding safety and to view available training resources, including helpful videos, visit <https://safeu.utah.edu>. To report suspicious activity or to request a courtesy escort, call campus police at (801) 585-2677 (COPS).

**Addressing Sexual Misconduct:** Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a Civil Rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status, or genetic information. If you or someone you know has been harassed or assaulted on the basis of your sex, including sexual orientation or gender identity/expression, you are encouraged to report it to the university officials in the Office of Equal Opportunity and Title IX, 383 S. University Street, (801) 581-8365, <https://oeo.utah.edu/contact-us/index.php>, or to the Office of the Dean of Students, 270 Union Building, (801) 581-7066. <https://deanofstudents.utah.edu/>. For support and confidential consultation, contact the Center for Campus Wellness, 350 Student Services Building, (801) 581-7776. To report to police, contact the Department of Public Safety, (801) 585-2677 (COPS), <https://police.utah.edu/>.

**Academic Misconduct:** It is expected that students comply with University of Utah policies regarding academic honesty, including but not limited to refraining from cheating, plagiarizing, misrepresenting one's work, and/or inappropriately collaborating. This includes the use of generative artificial intelligence (AI) tools without citation, documentation, or authorization. Students are expected to adhere to the prescribed professional and ethical standards of the profession/discipline for which they are preparing. Any student who engages in academic dishonesty or who violates the professional and ethical standards for their profession/discipline may be subject to academic sanctions as per the University of Utah's Student Code: Policy 6-410: Student Academic Performance, Academic Conduct, and Professional and Ethical Conduct.

Plagiarism and cheating are serious offenses and may be punished by failure on an individual assignment, and/or failure in the course. Academic misconduct, according to the University of Utah Student Code:

“... includes, but is not limited to, cheating, misrepresenting one’s work, inappropriately collaborating, plagiarism, and fabrication or falsification of information... It also includes facilitating academic misconduct by intentionally helping or attempting to help another to commit an act of academic misconduct.”

For details on plagiarism and other important course conduct issues, see the U’s Code of Student Rights and Responsibilities, <https://regulations.utah.edu/academics/6-400.php>